

§12.2 Vectors

Homework : 25,29,40

Vectors :

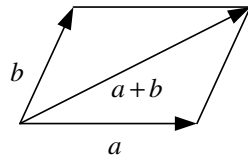
(1) 幾何(物理)觀點：有長度、方向的量

(2) 代數觀點： $\overrightarrow{op} = \langle a, b, c \rangle$ $P = (a, b, c)$.

$$\text{長度} = |\overrightarrow{op}| = \sqrt{a^2 + b^2 + c^2}$$

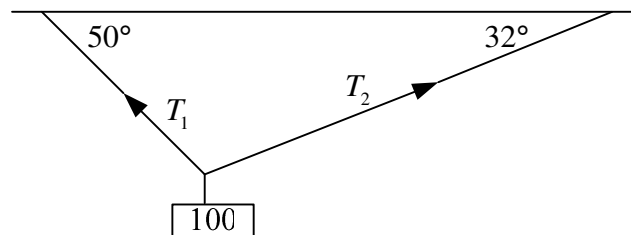
(3) 運算方式

幾何：



代數： $a = \langle a_1, a_2, a_3 \rangle$, $b = \langle b_1, b_2, b_3 \rangle$
 $\Rightarrow a \pm b = \langle a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3 \rangle$.

Example 1: A 100-lb weights hangs from two wires as shown below. Find the tension (forces) T_1 and T_2 in both wires and their magnitudes.



Solution :

$$\begin{aligned} |T_1| \sin 50^\circ + |T_2| \sin 32^\circ &= 0 \\ |T_1| \cos 50^\circ &= |T_2| \cos 32^\circ \end{aligned}$$

$$\begin{aligned} \Rightarrow T_1 &= \frac{100}{\sin 50^\circ + \tan 32^\circ \cos 50^\circ} \approx 85.64 lb \\ T_2 &= \frac{|T_1| \cos 50^\circ}{\cos 32^\circ} \approx 64.91 lb \end{aligned}$$

$$\begin{aligned} |T_1| &= (|T_1| \sin 50^\circ) j - (|T_1| \cos 50^\circ) i \approx -55.05 i + 65.60 j \\ |T_2| &= (|T_2| \sin 32^\circ) j + (|T_2| \cos 32^\circ) i \approx 55.05 i + 34.40 j \end{aligned}$$